

Determining Launch Parameters To Hit Targets

SYNOPSIS

This is a lab-companion worksheet, worth **10% bonus credit on the Module 2 exam**. Your launcher sits on the bench and fires a projectile at a fixed launch speed. Your job is to (1) find that launch speed two independent ways — a timed vertical launch and a photogate reading — and then (2) work the projectile-motion equations *backward*: instead of "where does it land?", you're asked "what angle makes it land *here*?" for targets set 1 m, 2 m, and 3 m from the launcher. That's called an **inverse problem** — solving for a condition (the angle) that produces a chosen outcome, rather than predicting an outcome from given conditions. As you'll see, a target is usually reachable by two different launch angles — a steep **high arc** and a flatter **low arc** — the **two-angle solution**.

PERFORMANCE EXPECTATIONS By the end of this note, you will be able to...

- PE1 Determine a launcher's launch speed two independent ways** — a timed vertical launch and a photogate reading — and compare them.
- PE2 Judge which velocity measurement is more reliable**, and explain the sources of error in each method.
- PE3 Set up the inverse projectile problem** — given launch height H , launch speed V_0 , and horizontal target distance D , solve for the launch angle θ .
- PE4 Recognize and explain the two-angle solution** — why a high arc and a low arc can both hit the same target.
- PE5 Apply the method** to find the launch angle(s) needed to hit targets at 1 m, 2 m, and 3 m from your bench-top launcher.



You already know this

You've already built the projectile-motion toolkit in N3 and N4. This note doesn't teach new physics — it asks you to run that toolkit *in reverse*, and to trust a measurement over a guess:



Free-throw vs. lob pass

You can sink the same hoop with a flat, fast shot or a high, arcing one. Same target, two different angles — that's the two-angle solution in action.



Working backward

N4 asked "given launch speed and angle, where does it land?" This note flips the question: "given where I want it to land, what angle do I need?" — same equation, solved for a different variable.



Stopwatch vs. sensor

You've compared a hand-timed measurement to an instrument reading before (like timing a lap by stopwatch vs. a timing gate). Human reaction time adds error that an electronic gate doesn't have.

In this note, we work through

- I. How can we find the launcher's launch speed two different ways — and which one should we trust more?
- II. Given the launch height, launch speed, and a target's horizontal distance, how do we solve for the launch angle?
- III. Why does the same target usually have *two* possible launch angles — a high arc and a low arc?
- IV. What angle(s) hit targets at 1 m, 2 m, and 3 m from the bench-top launcher?

Finding the launch speed, two ways

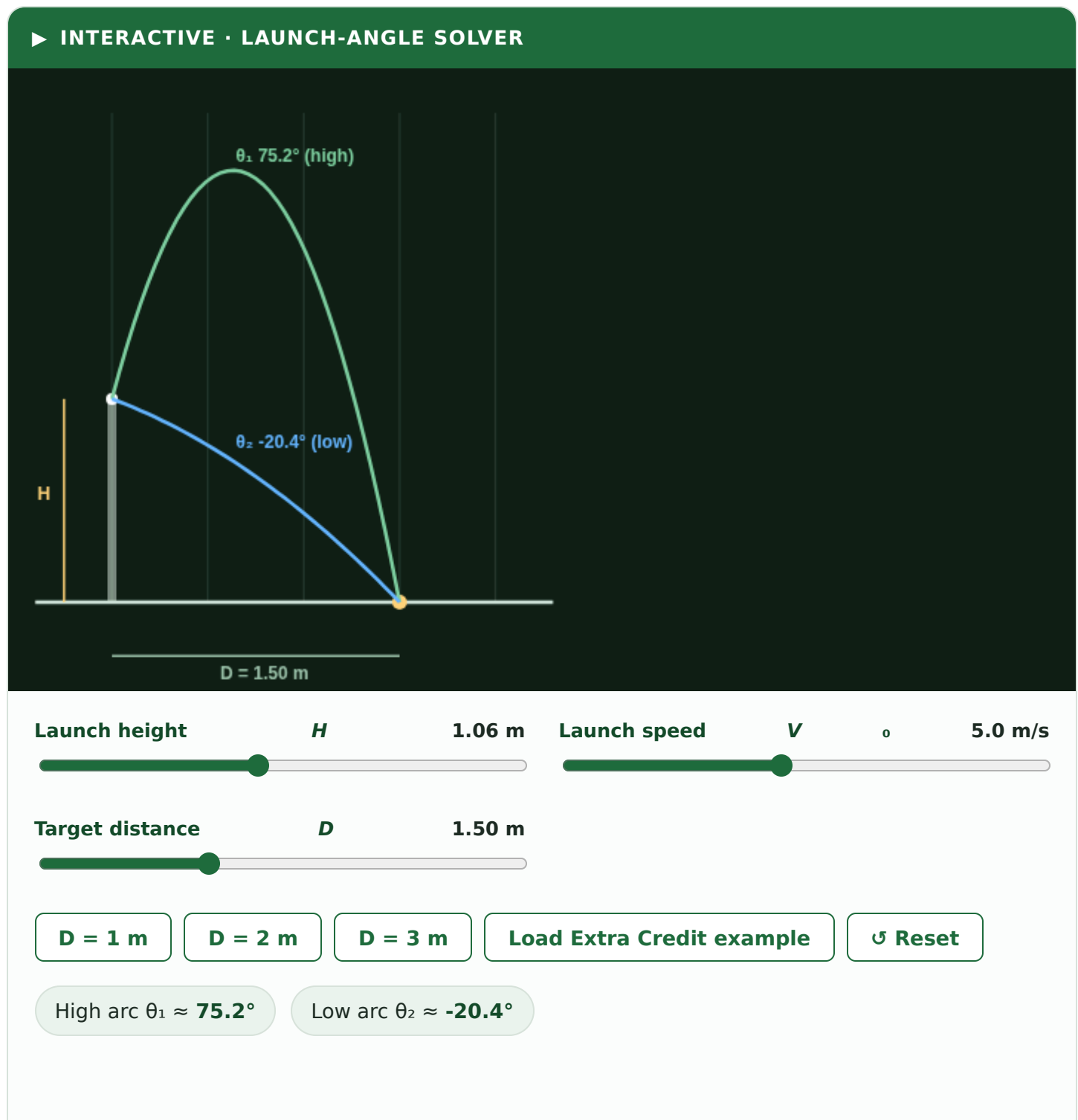
Method 1 — timed vertical launch. Point the launcher straight up and fire. Time the flight from launch to the moment it returns to launch height, t . By symmetry, the ball spends half that time going up and half coming down, so the launch speed is $V_0 = g \cdot t/2$. This method is limited by your reaction time on the stopwatch — twice, once at launch and once at return.

Method 2 — photogate. The launcher has a photogate right at the muzzle that times how long the projectile blocks a light beam of known width as it exits, giving V_0 directly, electronically, at the instant of launch. There's no human reaction time in this measurement at all — the main uncertainty is the gate's own timing resolution and the width measurement, both far smaller than a human stopwatch-press.

Record both values on your worksheet and compare them. In general, **the photogate reading is the more reliable of the two**, since it removes the two human reaction-time errors baked into the timed method.

Play with it: launch-angle solver

Set the launch height H , launch speed V_0 , and target distance D . The solver finds both launch angles — high arc and low arc — that land exactly on the target, the same way you'll solve for your own launcher's angles below.



Both trajectories start at the same launch point and land at exactly the same spot — they just get there differently.

Predict first: solving for a launch angle sometimes gives a *negative* answer, like $\theta = -20.4^\circ$.

What does that mean?

The math failed — throw that answer out.

The launcher must aim below the horizontal, since it starts above the target.

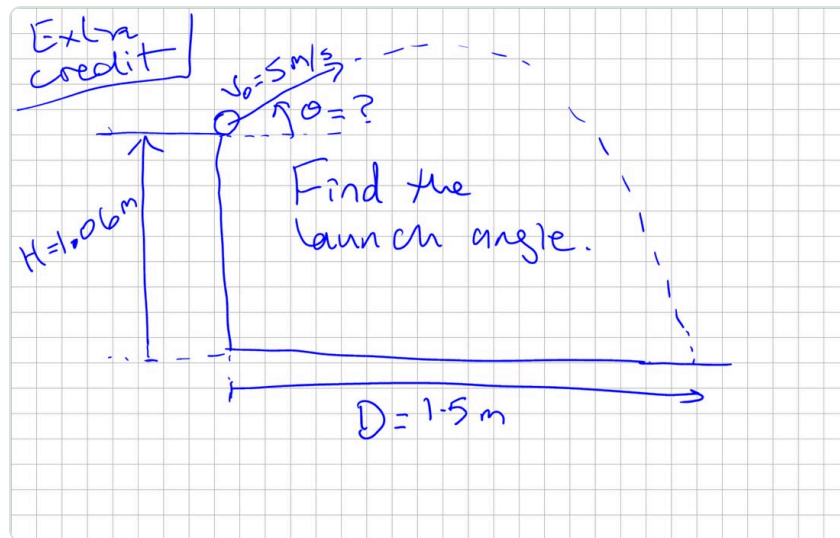
The projectile travels backward, away from the target.

You should retype the angle as a positive number and use it as-is.



Worked example (optional) — Extra Credit

Same quadratic N4 derives in general, worked here for specific numbers.



Dr. Cho's Extra Credit sketch — the launcher sits a height H above the bench where the target line is marked.

Given: launch height $H = 1.06$ m, launch speed $V_0 = 5$ m/s, target distance $D = 1.5$ m. Find the launch angle(s) θ .

1. Start from the two position equations, with the target level as $y = 0$ and the launch point at $y = H$:

$$x(t) = V_0 \cos \theta \cdot t, \quad y(t) = H + V_0 \sin \theta \cdot t - \frac{1}{2}gt^2.$$

2. Solve the x-equation for the time of flight: $t = D / (V_0 \cos \theta)$, and substitute into the y-equation, using $y = 0$ at the target.
3. Replace $1/\cos^2 \theta$ with $1 + \tan^2 \theta$ and collect terms. This is exactly the step N4 carries out in general — here it turns the y-equation into a **quadratic in $\tan \theta$** :
 $k \cdot \tan^2 \theta - D \cdot \tan \theta + (k - H) = 0$, where $k = gD^2 / (2V_0^2)$.
4. Plug in the numbers: $k = (9.8)(1.5)^2 / (2 \cdot 5^2) = 0.441$. The quadratic becomes $0.441 \cdot \tan^2 \theta - 1.5 \cdot \tan \theta - 0.619 = 0$.
5. Solve with the quadratic formula for $u = \tan \theta$. The two roots give the two launch angles.

Result: $\theta_1 = 75.2^\circ$ (high arc) and $\theta_2 = -20.4^\circ$ (low arc — aimed *below* the horizontal, since the launcher starts 1.06 m above the target line and still has room to fall).

N4 derives this same quadratic-in- $\tan \theta$ equation in general, for any launch height, speed, and target distance — this worked example just plugs in Dr. Cho's specific numbers. Try them yourself in the solver above (the "Load Extra Credit example" button sets it up for you).

Watch: working through the note

WALKTHROUGH VIDEO · ~8 MIN

Video embed goes here — print the lab worksheet below before you press play.

Exit ticket

Four quick questions — check yourself before lab.

Q1. Between the timed vertical launch and the photogate, which launch-speed measurement is generally more reliable, and why?

- ☐ The timed launch — a stopwatch is always more accurate than electronics.
- ☐ The photogate — it removes the human reaction-time error built into hand-timing.
- ☐ Neither — both are equally reliable in every case.
- ☐ The timed launch — it directly measures the angle too.

Q2. Why can two different launch angles hit the exact same target?

- ☐ They can't — a target has exactly one correct launch angle.
- ☐ A steep, lobbed arc and a flatter, faster-arriving arc can both cross the same landing point — one high, one low.
- ☐ Only if the launch speed changes between the two shots.
- ☐ Only if the target is exactly at the launch height.

Q3. Using the Extra Credit numbers ($H = 1.06$ m, $V_0 = 5$ m/s, $D = 1.5$ m), what is the high-arc launch angle θ_1 , to the nearest 0.1° ?

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Q4. In the equation $k \cdot \tan^2\theta - D \cdot \tan\theta + (k - H) = 0$, what does k stand for?

- ☐ $k = g D^2 / (2 V_0^2)$
- ☐ $k = V_0 / g$

☐ $k = H / D$

☐ k is just a made-up constant with no physical meaning.

Check my answers

Clear

Your notes

BEFORE LAB

Guided worksheet — blank (PDF)

Print this and bring it to the bench — you'll fill in your measured launch speed, height, and angles as you go.

AFTER LAB

Completed notes (PDF)

Released after the bonus-credit deadline — check your work against it.